

1 Grammar Rules

Types and Contexts

$$\begin{array}{ll}
\sigma, \tau & ::= \mathbf{unit} \mid \mathbf{num} \mid \sigma \otimes \sigma \mid \alpha \quad (\text{Types}) \\
\alpha & ::= m(\sigma) \quad (\text{Discrete Types}) \\
\Gamma & ::= \emptyset \mid \Gamma, x :_r \sigma \quad (\text{Linear Typing Contexts}) \\
\Phi & ::= \emptyset \mid \Phi, z : \alpha \quad (\text{Discrete Typing Contexts})
\end{array}$$

Expressions

$$\begin{array}{l}
e, f ::= x \mid z \mid () \mid !e \mid (e, f) \mid \mathbf{inl} \, e \mid \mathbf{inr} \, e \\
\quad \mid \mathbf{let} \, x = e \, \mathbf{in} \, f \mid \mathbf{let} \, (x, y) = e \, \mathbf{in} \, f \mid \mathbf{dlet} \, z = e \, \mathbf{in} \, f \mid \mathbf{dlet} \, (x, y) = e \, \mathbf{in} \, f \\
\quad \mid \mathbf{case} \, e' \, \mathbf{of} \, (\mathbf{inl} \, x.e \mid \mathbf{inr} \, y.f) \mid \mathbf{add} \, e \, f \mid \mathbf{sub} \, e \, f \mid \mathbf{mul} \, e \, f \mid \mathbf{dmul} \, e \, f \mid \mathbf{div} \, e \, f
\end{array}$$

2 Typing Rules

Context Variables

$$\frac{}{\Phi \mid \Gamma, x :_r \sigma \vdash x : \sigma} \text{(Var)} \qquad \frac{}{\Phi, z : \alpha \mid \Gamma \vdash z : \alpha} \text{(DVar)}$$

Tensor Destructors

$$\frac{\Phi \mid \Gamma \vdash e : \tau_1 \otimes \tau_2 \quad \Phi \mid \Delta, x :_r \tau_1, y :_r \tau_2 \vdash f : \sigma}{\Phi \mid r + \Gamma, \Delta \vdash \mathbf{let} \, (x, y) = e \, \mathbf{in} \, f : \sigma} \text{(\otimes E}_{\sigma}\text{)}$$

$$\frac{\Phi \mid \Gamma \vdash e : \alpha_1 \otimes \alpha_2 \quad \Phi, z_1 : \alpha_1, z_2 : \alpha_2 \mid \Delta \vdash f : \sigma}{\Phi \mid \Gamma, \Delta \vdash \mathbf{dlet} \, (z_1, z_2) = e \, \mathbf{in} \, f : \sigma} \text{(\otimes E}_{\alpha}\text{)}$$

Let Bindings

$$\frac{\Phi \mid \Gamma \vdash e : \tau \quad \Phi \mid \Delta, x :_r \tau \vdash f : \sigma}{\Phi \mid r + \Gamma, \Delta \vdash \mathbf{let} \, x = e \, \mathbf{in} \, f : \sigma} \text{(Let)} \qquad \frac{\Phi \mid \Gamma \vdash e : \alpha \quad \Phi, z : \alpha \mid \Delta \vdash f : \sigma}{\Phi \mid \Gamma, \Delta \vdash \mathbf{dlet} \, z = e \, \mathbf{in} \, f : \sigma} \text{(DLet)}$$

Bang Destructor

$$\frac{\Phi \mid \Gamma \vdash e : \sigma}{\Phi \mid \Gamma \vdash !e : m(\sigma)} \text{(Disc)}$$

Operations.

$$\frac{}{\Phi \mid \Gamma, x :_{\varepsilon+r_1} \mathbf{num}, y :_{\varepsilon+r_2} \mathbf{num} \vdash \{\mathbf{add}, \mathbf{sub}\} \, x \, y : \mathbf{num}} \text{(Add, Sub)}$$

$$\frac{}{\Phi \mid \Gamma, x :_{\varepsilon/2+r_1} \mathbf{num}, y :_{\varepsilon/2+r_2} \mathbf{num} \vdash \mathbf{mul} \, x \, y : \mathbf{num}} \text{(Mul)}$$

$$\frac{}{\Phi \mid \Gamma, x :_{\varepsilon/2+r_1} \mathbf{num}, y :_{\varepsilon/2+r_2} \mathbf{num} \vdash \mathbf{div} \, x \, y : \mathbf{num} + \mathbf{unit}} \text{(Div)}$$

$$\frac{}{\Phi, z : m(\mathbf{num}) \mid \Gamma, x :_{\varepsilon+r} \mathbf{num} \vdash \mathbf{dmul} \, z \, x : \mathbf{num}} \text{(DMul)}$$